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Phase-2 Mid-Term Evaluation • October 2026

KAVACH: Coverage of Project Till Now vs. Evaluation Criteria & Course Outcomes

Comprehensive Mid-Term Progress Audit Across Rubrics C1–C10 & Course Outcomes CO1 (Multi-Agent), CO2 (Security/Privacy), CO3 (Multimodal/Tasks)

Rubric / ID	Category / Syllabus Module	Mapped CO	Planned Target Benchmark	Measured Implementation Reality	Verification Suite	Status
C1 / CRS-01	Problem Definition & Threat Model	CO1, CO2	USENIX 2025 Slopsquatting & DPDP PII leakage model	Formally expanded to Banking Memo Injection (OWASP LLM01) & DPDP Act	01_UPDATED_PROBLEM.md	100% MET
C2 / CRS-02	Objectives & Measurable Outcomes	CO1, CO2, CO3	Quantitative milestones aligned with CO1, CO2, CO3	142 endpoints, 0.017ms guardrail, 10-stage FSM, 300 test assertions	docs/METRICS.json	100% MET
C3 / CRS-03	Agentic RAG & Vector Embeddings	CO1, CO2	AST code chunking + dense vector similarity search	Qdrant vector store with 384-d MiniLM embeddings; AST syntax chunker	test_rag.py (21/21 PASS)	100% MET
C4 / CRS-04	Multi-Agent Orchestration & FSM	CO1	Finite State Machine with sub-agent delegation	10-Stage FSM in orchestrator.py, stateful Crew Flows (@start, @listen)	test_kavach_100_suite.py	100% MET
C5 / CRS-05	Google ADK 8-Hook Callbacks	CO1, CO2	Google ADK lifecycle states & medical triage	Native in-repo Google ADK 8-hook callbacks (callbacks.py) + medical triage	test_adk_triage.py (17/17)	100% MET
C6 / CRS-06	Progress vs Plan & Slip Analysis	CO1, CO2, CO3	16-Week schedule audit with slip mitigations	100% Phase-2 milestones met; documented VRAM & lock slip mitigations	02_UPDATED_PLAN.md	100% MET
C7 / CRS-07	Technical Understanding & Math	CO1, CO2	Formulations in entropy, graph theory, vectors	Shannon entropy H>4.5, AST reachability closure B(v), Verhoeff checksum	secret_detector.py	100% MET
C8 / CRS-08	Problem Solving & Bounded ReAct	CO2, CO3	Autonomous self-healing without infinite loops	Subprocess pytest sandbox, 5.0s watchdog timer, bounded reflection N<=3	focused_check.py (5/5 PASS)	100% MET
C9 / CRS-09	Working Documentation & SSOT	CO1, CO2, CO3	OpenAPI specs, single source of truth, reports	Master METRICS.json, Swagger docs at /docs, 86KB synopsis report	docs/METRICS.json	100% MET
C10 / CRS-10	Team Responsibility & Rotation	CO1, CO2, CO3	Systematic role rotation across project phases	RACIS matrix tracking Dhruv (Arch), Hemant (Security), Rajat (Telemetry)	TEAM_MATRIX.md	100% MET
EXP-01	Supply-Chain Slopsquatting Firewall	CO2	Prevent execution of hallucinated third-party libraries	AST import extractor + real-time PyPI JSON registry query with LRU cache	package_verifier.py	100% MET
EXP-02	Bharat Core Banking & SAGA Escrow	CO3	Financial transaction governance with RBI compliance	15 banking routes at /bank, SAGA distributed rollbacks, >₹50,000 Escrow	test_banking_100_suite.py	100% MET